

SHUBHAM GAUR

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SDET/ QA Automation Engineer

Software Development Engineer in Test (SDET) with 5 years of experience in building scalable automation frameworks and ML-assisted validation techniques for enterprise software. Hands-on experience in test strategy design, framework architecture and development and AI/ML integration to analyze failures and enhance quality under real-world constraints, including a multilingual OCR+NLP evaluation system, reducing defect escape rate from 15% to 3%. Strong foundation in C#, Python and modern test tools like Selenium, Playwright and Postman/ RestAssured. Adept at Agile testing, CI/CD Integration and Distributed System Testing with growing capabilities in applying machine learning concepts to improve quality and drive innovation in automation practices.

SKILLS

- **Automation & Testing:** Test Automation Framework Design (Keyword/Data-driven), API Testing (REST APIs), Database Validation, Functional & Non-Functional Testing, End-to-End Testing
- **Languages:** C#, Python, SQL
- **Tools & Technologies:** Selenium, Playwright, NUnit, MSTest, PyTest, Postman, RestAssured, Azure DevOps
- **Specialization:** AI/ML in Test Automation, GenAI (Prompt Engineering), Optical Character Recognition (OCR) based Validation
- **Methodologies:** Shift-Left Testing, Test Strategy Design, Test-Driven Development (TDD), Behavior-Driven Development (BDD), Quality Metrics, Root Cause Analysis
- **Software Engineering:** Agile/Scrum, CI/CD (Azure Devops), Data Structures & Algorithms, Distributed System Validation, Git, OOPs, Rest APIs, Automation System Architecture
- **Soft Skills:** Analytical Problem-Solving, Cross-Functional Collaboration, Mentorship & Team Development, Stakeholder Communication

PROJECTS

ML-Powered Localization Automation Framework

HCLTech | 2024-25

- Designed and developed UI content validation framework to automate translation quality assessment across 12+ languages.
- Architected an end-to-end system: screenshot capture -> OCR Test extraction -> Validation
- Conducted comparative evaluation of OCR engines across 8 languages, analyzing error distributions under varying image noise.
- Reduced translation failure rate by 25%, catching critical errors missed by manual review.

Local RAG System for Private Document Q&A

Personal Project | 2025

- Designed and implemented an end-to-end Retrieval-Augmented Generation (RAG) system for

answering queries over private documents without external API dependencies.

- Built full pipeline: document ingestion, token-based chunking with overlap, embedding generation, vector retrieval and LLM-conditioned response generation.
- Gained practical understanding of text embeddings, retrieval accuracy, evaluation of generated response.

WORK EXPERIENCE

Software Quality Engineer

Becton Dickinson | June 2026 - Present

- Improved automation framework performance by 40%, reducing regression execution time from 10 hours to 6 hours through framework optimization and code refactoring.
- Eliminated flaky test cases, improved automation stability, and enhanced code quality using AI-assisted development tools, resulting in higher test reliability and reduced defect leakage.

Senior Test Engineer

HCLTech (formerly Becton Dickinson) | Aug 2023 - June 2026

- Transitioned from traditional automation into ML-assisted evaluation systems using OCR and NLP for multilingual content validation, reducing production quality failure by 70%.
- Built an ML-based localization testing solution, integrating Optical Character Recognition (OCR) for screenshot text extraction and multilingual validation, significantly reducing defect leakage and improving translation accuracy across 12+ languages.
- Developed hybrid and keyword-driven test automation frameworks using C#, Python, Selenium WebDriver and Playwright, with integrated Extent Reports, laying a scalable foundation adopted across multiple projects.
- Integrated automation suites into Azure DevOps CI/CD pipelines, enabling continuous testing and centralized reporting of automation health.
- Developed test reporting dashboards to track automation coverage, defect trends and release readiness.
- Performed API and backend validations for JSON Schema, request/response payloads for distributed services, ensuring functional correctness.
- Mentored junior engineers on automation best practices, conducted code reviews, and enforced Git- based version control best practices.

Test Engineer

Coforge | March 2021 - July 2023

- Designed C#-based automation workflows incorporating Excel-driven data inputs and Extent Reports-based execution dashboards to enhance visibility and reduce manual testing effort.
- Performed API verification for automobile system services, including financial and billing modules, ensuring request/response correctness and schema adherence.
- Performed database validations using SQL to ensure integrity across systems.
- Validated complex business flows (pricing, invoicing, calculations)
- Conducted root cause analysis on failures, collaborating with development teams to identify data and integration issues.

EDUCATION

Bachelor's in Computer Science Engineering | Chandigarh University, Gharuan

Session: 2017-2021 | 7.86 CGPA

12th (CBSE) | Kendriya Vidyalaya, Chandigarh

Session: 2016-2017 | 87%

10th (CBSE) | Kendriya Vidyalaya, Chandigarh

Session: 2014-2015 | 10 CGPA

CERTIFICATIONS

Core Python Certification

01/2026 - 06/2026 | ATL Education Foundation (6-week program)